Action recognition

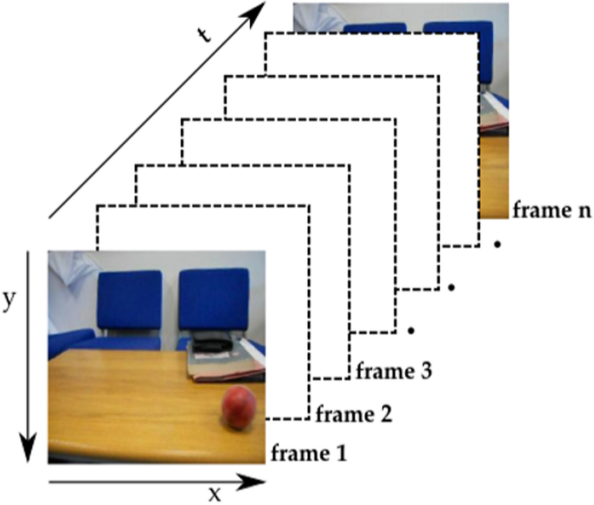
Laboratory of robotics and control systems
ITT PROJECT MANAGEMENT SERVICES L.L.C

Plan

1. About video
2. Optical flow
3. Action recognition

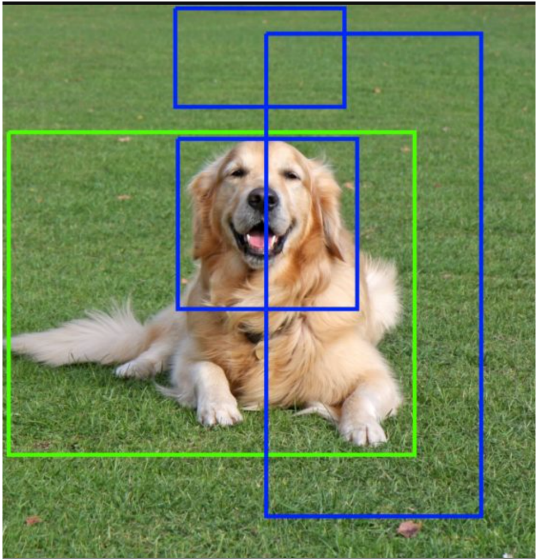
Stream vs sequence

- **Online**: Process video as a stream, handling one image at a time at ≈ 15 FPS.
- **Offline**: Treat video as a fixed sequence of frames, where FPS is irrelevant.



Attitude to model's prediction

- $\#false\ positives$ is multiplied by frames per second
- $\#false\ positives$ should be adequate
- Need models with extremely high precision (> 0.99)
- Have to overfit models to exact shooting scenarios



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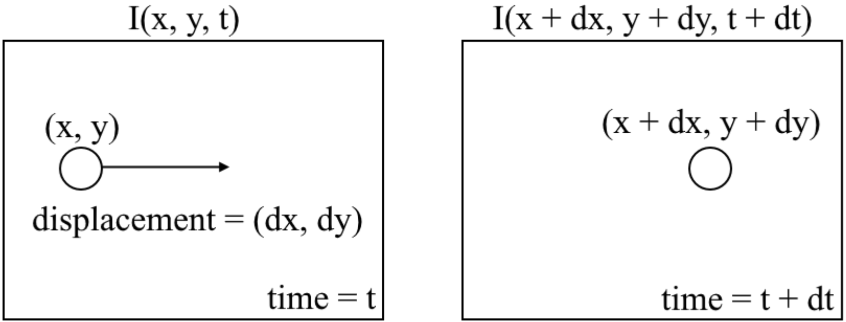
What is optical flow



Optical flow is an estimate of the **visible movement of pixels**

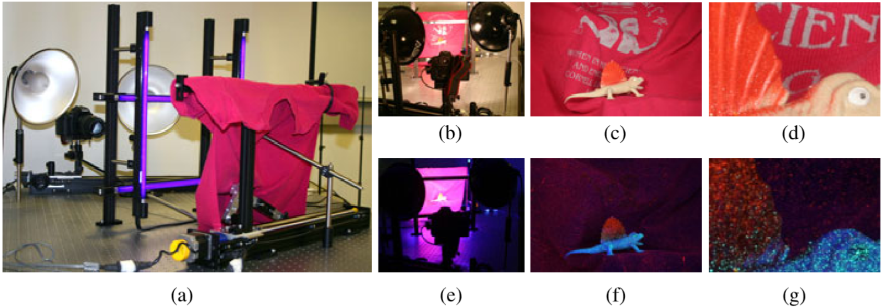
Dense optical flow

- Estimates the displacement of each pixel
- Direction of vector is encoded with Hue in HSV
- Length of vector is encoded with in HSV

Sparse optical flow

- Estimates the motion of a given set of points
- Visualization by tracking these points
- Significant savings in computing resources

Middlebury dataset



- 2 sets of images: under normal and UV light
- Objects are coated with fluorescent material that glows under UV light
- Every pixel in the image has a distinct intensity or color
- Optical flow is computed by comparing texture positions before and after motion

KITTI dataset

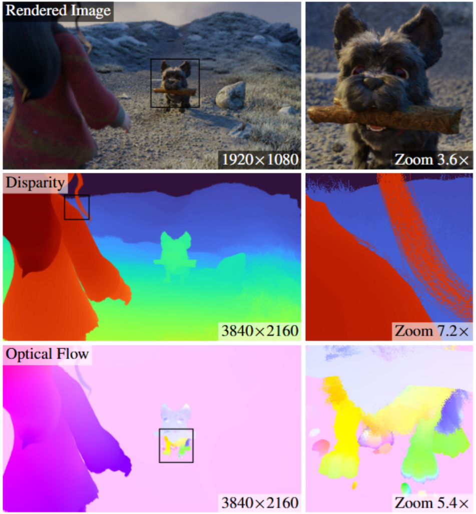
- IMU/GPS estimates motion of shooting vehicle
- Lidar helps to segment moving objects (cars)
- 3D CAD models interpolates sparse Lidar output
- 3D velocities are projected into image plane
- Stereo vision estimates background optical flow

Sintel: rendered cartoon

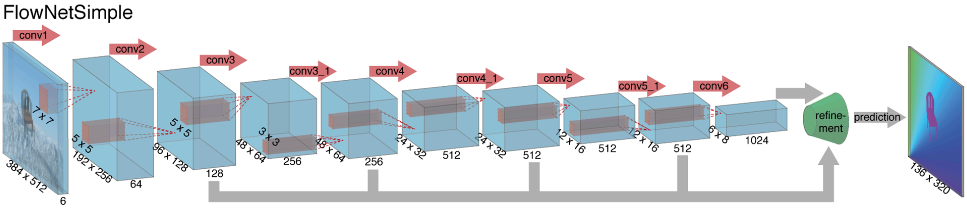
- All objects and backgrounds are created in Blender
- Per-object motion vectors are known by design
- 1.5K images, 1024 x 436 resolution

Spring: higher resolution

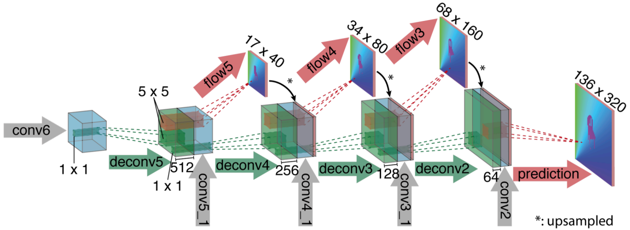
- 6000 FullHD images
- Ground truth in 4K resolution
- Subpixel accuracy for thin details



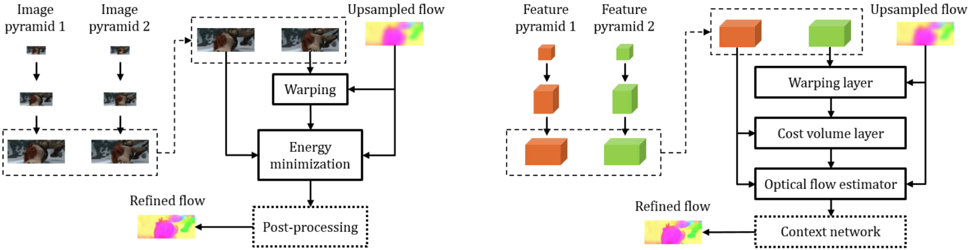
FlowNet (2015)



- Concatenate 2 sequential images
- deconv=unpool + conv
- loss = MSE (or endpoint error 'EPE')

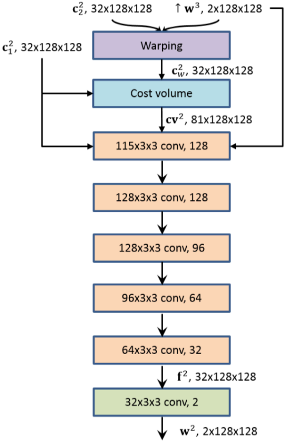


PWC-Net (2018)

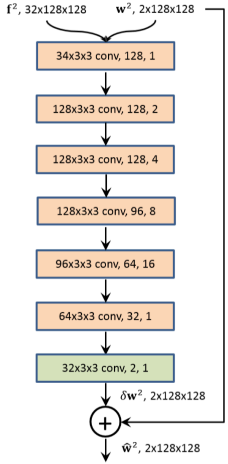


- Image pyramid — just downsampled images
- Upsampled flow at level l — upsampled flow from level $l - 1$, where $l = 1$ corrspond to the smallest image pair
- Warping—shifting pixels of image (features) in the direction from upsampled flow
- Cost volume layer – compute cosine similarity between vector from $featuremap_1$ and vectors from $featuremap_2$ in $d \times d$ area (cost volume tensor shape: $d^2 \times H \times W$)

PWC-Net (2018) part 2

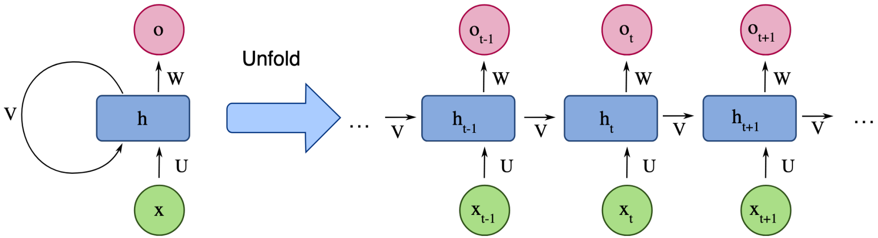


Optical flow estimator at level 2



Context network at level 2

Recurrent Neural Network (RNN)



$$h_t = \tanh(Vh_{t-1} + Ux_t + b_h)$$

$$o_t = Wh_t + b_o$$

h_t : Hidden state at time step t

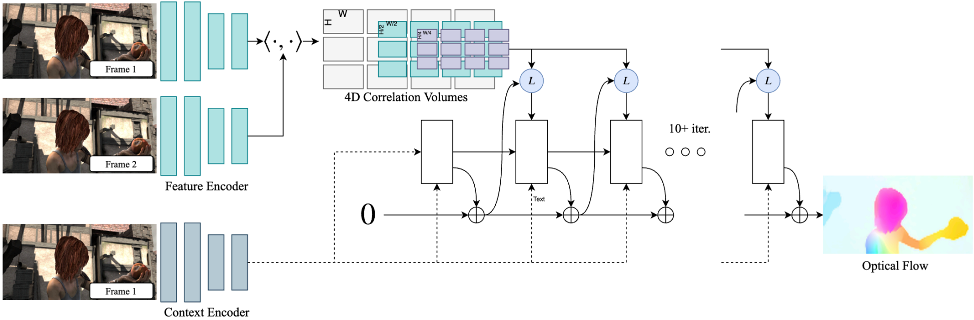
x_t : Input vector at time step t

o_t : Output vector at time step t

V, U, W : Trainable weight matrices

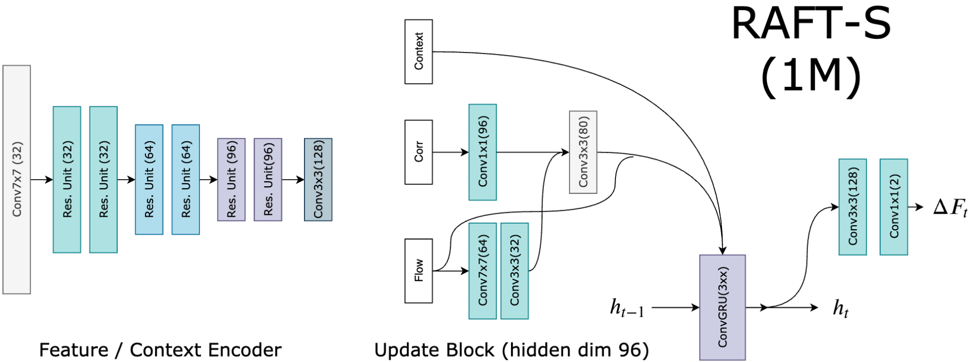
b_h, b_o : Trainable bias vector

RAFT (2020)



- Feature encoder (pairs of sequential frames), Context encoder (only first frame)
- Set of correlation (cost volume) tensors are extracted in different resolution after feature encoder
- Rnn predicts Δf: flow refinement
- L - extracts small grid (4×4) of features for each pixel features

RAFT (2020) part 2



- Backbone outputs features in 1/8 scale
- Input to rnn - concatenation of predicted flow, context and correlation
- Output flow is upsampled to original shape

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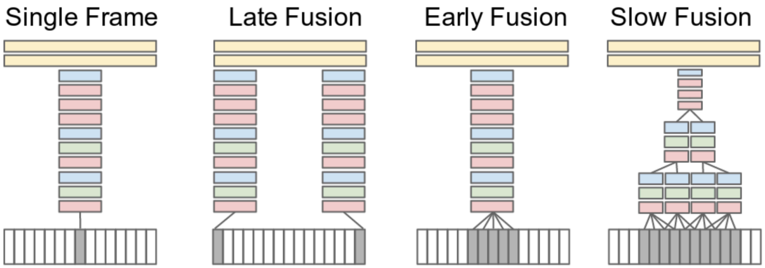
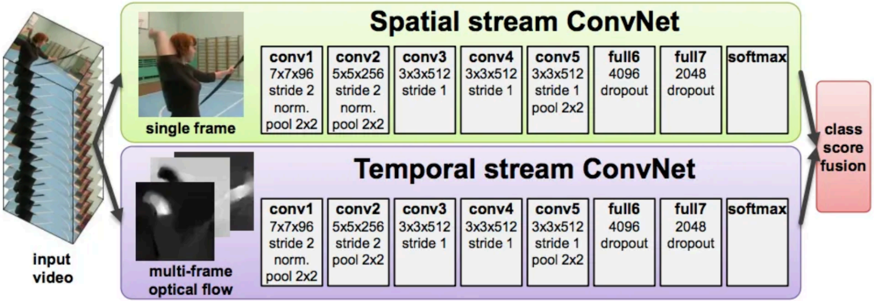
Video classification approaches



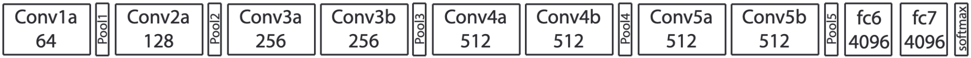
- **Single-frame**: only middle frame is processed by a 2D CNN (baseline)
- **Early Fusion**: process T consequtive frames with filter of size $11 \times 11 \times 3 \times T$
- **Late Fusion**: process 2 frames at long distance, fuse features before classification
- **Slow Fusion**: iteratively do temporal convolutions for small T

Two-Stream CNN (2014)



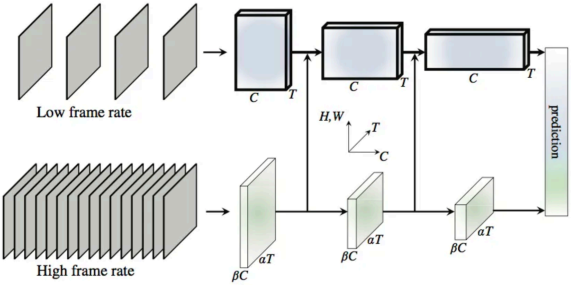
- Spatial stream uses only central frame
- Spatial stream may be pre-trained on ImageNet
- Temporal stream uses optical flow
- Optical flows from several consective frames are used
- SVM predicts final score

C3D (2014)



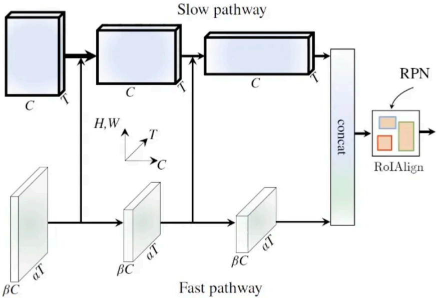
- Input tensor shape: (Batch, Channels, Depth (Frames), Height, Width)
- Allows the model to capture both spatial features (appearance) and temporal features (motion patterns) simultaneously

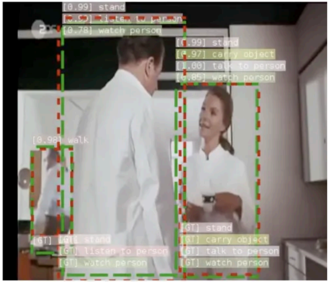
SlowFast (2018)



- **Slow pathway**: only 1/8th frames. Focused to capture contextual information or structure.
- **Fast pathway**: all the frames. Focused on motion determination.

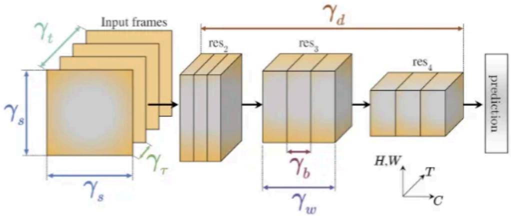
SlowFast: AVA action detection





- Person detection happens first (for example Faster R-CNN) with 1 FPS
- ROI pooling is applied to extract features only from detected human bounding boxes
- Final action labels are assigned per detected person per keyframe

X3D: Expanding Architectures (2020)

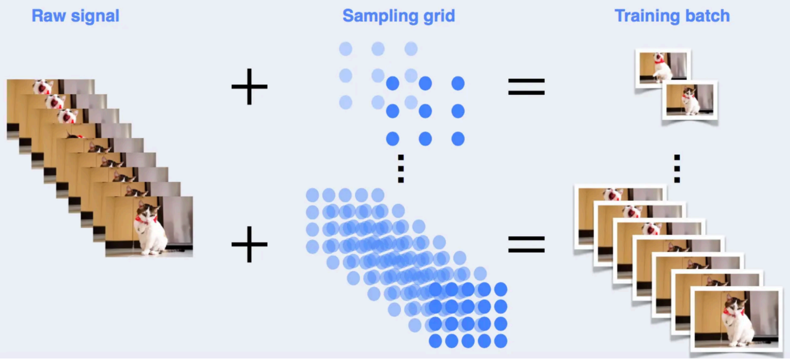


- Expand *one single axis* at a time such that computation is increased (e.g. ×2)
- Train and validate model and select the axis that achieves the best computation/accuracy trade-off

Expansion operations:

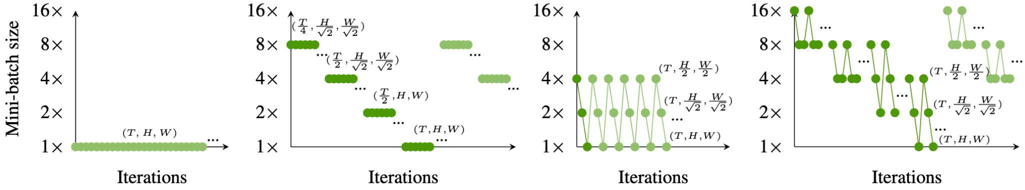
- *X-Fast* expands the temporal activation size γ_t by increasing the frame-rate, $1/\gamma_\tau$, and therefore temporal resolution, while holding the clip duration constant.
- *X-Temporal* expands the temporal size γ_t by sampling a longer temporal clip and increasing the frame-rate $1/\gamma_\tau$, to expand both duration *and* temporal resolution.
- *X-Spatial* expands the spatial resolution γ_s by increasing the spatial sampling resolution of the input video.
- *X-Depth* expands the depth of the network by increasing the number of layers per residual stage by γ_d times.
- *X-Width* uniformly expands the channel number for all layers by a global width expansion factor γ_w.
- *X-Bottleneck* expands the inner channel width of the center convolutional filter in each residual block.

Multigrid training



- Training at low spatial resolutions increase the speed of training
- With high resolution accuracy is higher
- Can we get benefits of both low resolution and higher resolution?

Baseline vs Multigrid



(a) **Baseline** (b) **Long cycles** (c) **Short cycles** (d) **Long + short cycles**

- **Long cycle**: trains really quickly in the start and then gets refined with a higher resolution to achieve maximum accuracy
- **Short cycle**: focus on mixture of multiple shapes

Thank you for your time

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